ICT
ASSIGNMENT 1

Object-Oriented Design of a Console E-Commerce Application

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1. Introduction

Information and Communication Technology (ICT) covers the digital tools and systems used to create, process, store, and share information. This report applies core ICT and object-oriented programming principles to a console-based e-commerce application developed in Java.

The application represents users, products, and customer orders. Separate customer and employee menus support different tasks, while classes, inheritance, encapsulation, and ArrayList collections organize the program's data and behavior.

2. Project Objective

The goal is to build a small e-commerce system that demonstrates class structure, relationships between objects, inheritance, data protection, and collection handling. Users can sign in, customers can create orders, and employees can maintain user and product records.

3. Requirements Analysis

The system requirements fall into three areas: information models, store operations, and console interaction. The main classes and their responsibilities are summarized below.

Class	Role in the system	Key information / behavior
User	Base class for customers and employees	ID, name, email, password, role
Customer	Represents a buyer	User data; creates orders
Employee	Represents a store employee	User data; manages records
Product	Base class for products	ID, name, price, stock
Book	Book product subtype	Author and ISBN
Electronics	Electronic product subtype	Warranty period
Clothing	Clothing product subtype	Size and color
OrderItem	A line within an order	Product reference and quantity
Order	A customer purchase	Order ID, customer, items, total
Store	Coordinates application data	Users, products, and orders
Main	Program entry point	Input, login, and menus

3.1 Functional requirements

• Authenticate users through email and password.

• Show products and user records in the console.

• Let employees add users and products.

• Allow customers to place orders containing multiple products and quantities.

• Check available stock before confirming an order.

• Calculate the total and print a receipt.

• Update stock after a completed purchase.

• Let employees inspect all orders and customers view their own orders.

3.2 Validation and limitations

Validation covers repeated user IDs, email addresses and product IDs, invalid menu choices, quantities that are zero or negative, and orders exceeding available stock. Data is kept in in-memory ArrayList collections; it is not retained when the application exits.

4. Object Modeling and Relationships

Customer and Employee inherit shared fields and behavior from User. Book, Electronics, and Clothing extend Product, so common product information can be implemented once and reused. An Order belongs to a Customer and contains OrderItem entries; each entry points to one Product and stores the requested quantity.

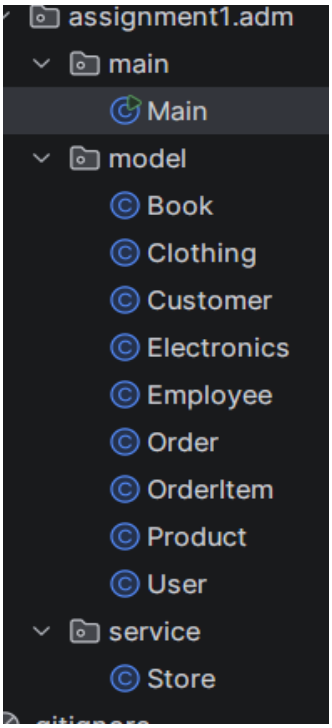
This structure supports orders with several different products and allows the same product to appear in multiple orders. Store maintains the collections of users, products, and orders and coordinates the main operations.

5. UML Class Diagram

The UML diagram on the next page summarizes the main classes, their inheritance links, and their associations. Getters and setters are omitted from the diagram to keep it readable; model attributes are accessed through controlled methods.

6. Design and Implementation

The source is organized under the root package **assignment1.adm**. The project separates model classes, store services, and the main application entry point.



6.1 Package organization

The model package contains User, Customer, Employee, Product, Book, Electronics, Clothing, OrderItem, and Order. Store is located in the service package, while Main is in the main package.

6.2 Encapsulation

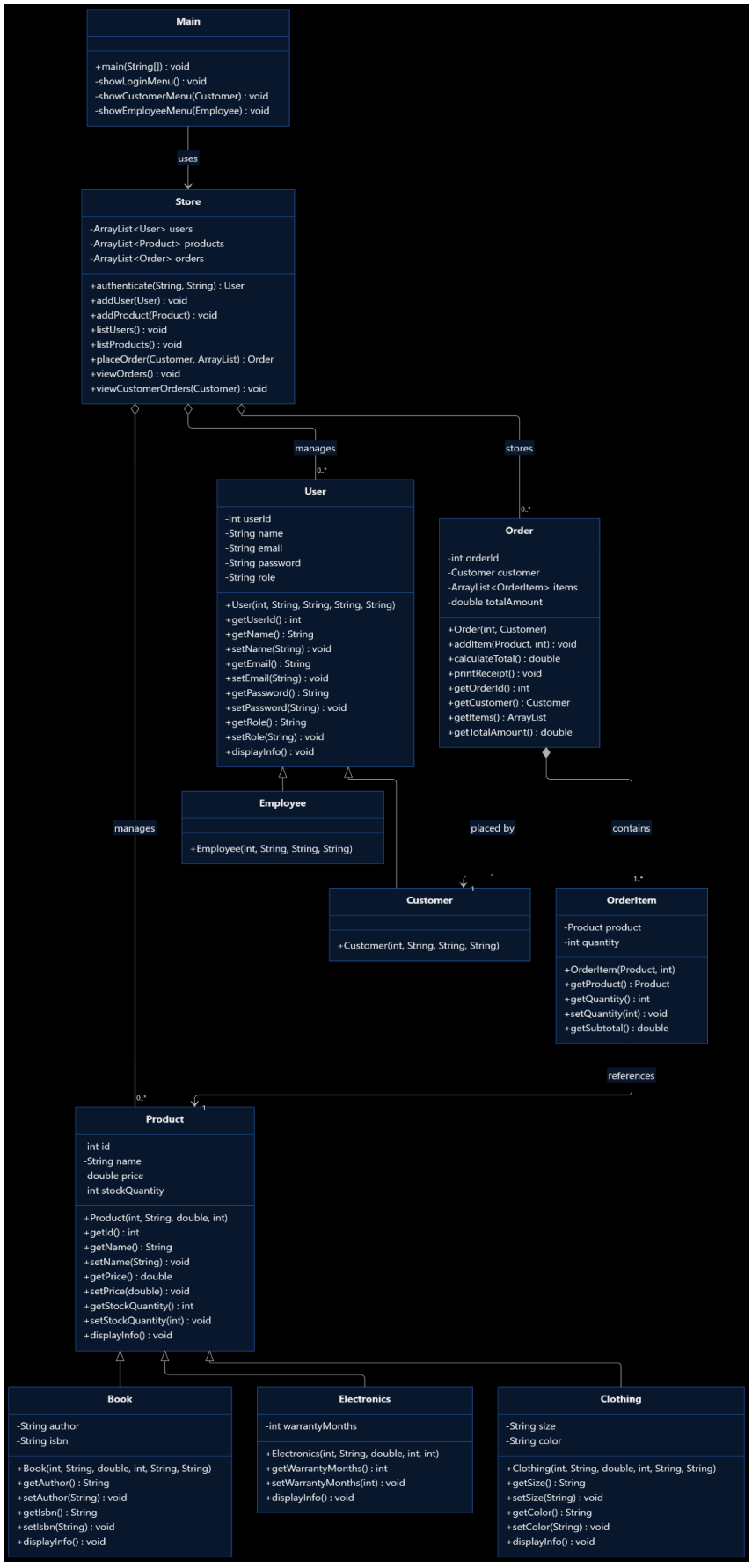
Model attributes are private. Getters provide read access, and setters are used when changes are allowed. Validation prevents negative prices, stock amounts, and warranty durations. Passwords are excluded from user-information output.

6.3 Inheritance and overriding

Customer and Employee reuse User functionality. The three product subclasses reuse Product data and override displayInfo() to show their own additional details.

UML CLASS DIAGRAM

Class structure and relationships



7. Conclusion

This project illustrates how object-oriented programming can be used to organize a basic e-commerce application. Inheritance represents specialized user and product types, encapsulation protects object attributes, and ArrayList collections hold the data used during a session.

Order and OrderItem model purchases that contain multiple products and quantities. Store coordinates validation and stock changes, while the console menus provide separate workflows for customers and employees.

The implementation addresses the main educational requirements for a Java console application. Its key limitation is that information exists only in memory and is lost after the program stops.

8. Source Code

Project repository:

https://github.com/ned667/ict-assignment1-dinmukhammed

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